

Q2-5

Levenspiel plot screencast:

1. Clear step by step approach to problem solving without skipping steps. Short video helps in maintaining students' focus.
2. Clear and to the point explanation, handwriting and background could be improved.

P2-2A

1. Revisit the data in **Table 2-1 Raw Data** and calculate the batch reactor (BR) times to achieve 10%, 50%, and 80% conversion when 100 moles of A are charged to a 400 dm³ reactor.
2. Revisit **Examples 2-1** through **2-3**. How would your answers change if the flow rate, F_{A0} , were cut in half? If it were doubled? What conversion can be achieved in a 4.5 m³ PFR and in a 4.5 m³ CSTR?
3. Revisit **Example 2-2**. Being a company about to go bankrupt, you can only afford a 2.5 m³ CSTR. What conversion can you achieve?
4. Revisit **Example 2-3**. What conversion could you achieve if you could convince your boss, Dr. Pennypincher, to spend more money to buy a 1.0 m³ PFR to attach to a 2.40 CSTR?
5. Revisit **Example 2-4**. How would your answers change if the two CSTRs (one 0.82 m³ and the other 3.2 m³) were placed in parallel with the flow, F_{A0} , divided equally between the reactors?
6. Revisit **Example 2-5**. (1) What is the worst possible way to arrange the two CSTRs and one PFR? (2) What would be the reactor volumes if the two intermediate conversions were changed to 20% and 50%, respectively? (3) What would be the conversions, X_1 , X_2 , and X_3 , if all the reactors had the same volume of 100 dm³ and were placed in the same order?
7. Revisit **Example 2-6**. If the term $C_{A0} \int_0^X \frac{dX}{-r_A}$ is 2 seconds for 80% conversion, how much fluid (m³/min) can you process in a 3 m³ reactor?

① BR design equation:

$$N_{A0} \frac{dX}{dt} = -r_A V$$

to solve for time needed to achieve certain conversion:

$$\int dt = \int N_{A0} \cdot \frac{dX}{-r_A V}$$

$$\Rightarrow \tau = N_{A0} \int_0^X \frac{dX}{-r_A V}$$

$$, N_{A0} = 100 \text{ mol A}$$

$$V = 400 \text{ dm}^3 (L)$$

$$x=0.1 \quad t = 100 \int_0^{0.1} \frac{dx}{-r_A(x) \cdot 0.4}$$

$$t = \frac{100}{0.4} \int_0^{0.1} \frac{dx}{-r_A(x)}$$

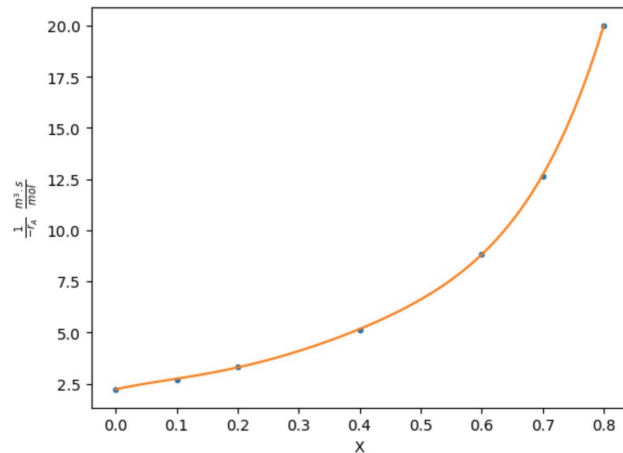
$$400 \cdot \frac{\text{m}^3 \cdot \text{s}}{\text{mol} \cdot \text{dm}^3} \times \left(\frac{10 \text{ dm}^3}{1 \text{ m}^3} \right)^3$$

$$400 \text{ dm}^3 \times \frac{1 \text{ m}^3}{(10 \text{ dm}^3)^3} = 0.4 \text{ m}^3$$

using python code provided
and modifying to solve for t:

```
Leven_Pfit = np.poly1d(np.polyfit(X, 1/-rA, 5))
```

```
Xp = np.linspace(0, 0.8, 100)
plt.plot(X, 1/-rA, '.', Xp, Leven_Pfit(Xp), '-')
plt.xlabel('X')
plt.ylabel(r"$\frac{1}{-r_A}$");
```



```
] # find t for a given conversion for BR
Xin = 0.0 # conversion at the inlet
Xout = 0.8
NA0overV = 250 # [mol/m3]
t = NA0overV*quad(Leven_Pfit, Xin, Xout)[0] #time to reach given conversion
tmin=t/60
print('t =', tmin, 'min')
```

t = 22.38655714473249 min

at $x=0.8 \quad t = 22.39 \text{ min}$

Same code used for $x=0.5$ & $x=0.1$

at $x \quad t = \frac{NA_0}{V(\text{m}^3)} \int_0^x \frac{dX}{-r_A(x)} \quad (\text{s})$

$x=0.5 \quad t = 8.17 \text{ min}$

$x=0.1 \quad t = 1.04 \text{ min}$

② for $FA_0 = 0.4 \text{ mol/s}$ if $FA_0 = 0.2 \text{ mol/s}$

CSTR volume for $x=0.8 \Rightarrow V = 6.4 \text{ m}^3$ $\Rightarrow V = 3.2 \text{ m}^3$

PFR $\gg \gg \Rightarrow V = 2.2 \text{ m}^3$ $\Rightarrow V = 1.1 \text{ m}^3$

$\Rightarrow$ if FA_0 is cut in half so is the volume of the reactor
if FA_0 is doubled $\Rightarrow$ volume also doubles

CSTR $V = 4.5 \text{ m}^3$

$$\Rightarrow X = 0.74 \\ \approx 74\%$$

PFR $V = 4.5 \text{ m}^3$

$$\Rightarrow X = 0.98 \\ \approx 98\%$$

* Answers obtained from python code provided

③

$V = 2.5 \text{ m}^3$ CSTR $X?$

```
# find CSTR conversion for a given volume
V = 2.5 # volume in m3
Xin = 0 # conversion at the inlet
def CSTR_mol_bal(X):
    return V - (X - Xin) * Leven_Pfit(X)

X = fsolve(CSTR_mol_bal, 0.5)

print('Xout =', X[0])

Xout = 0.6337706479068278
```

P2-6B Read the Web Module "Chemical Reaction Engineering of Hippopotamus Stomach" on the CRE Web site (http://www.umich.edu/~elements/6e/web_mod/hippo/index.htm).



1. Write five sentences summarizing what you learned from the Web Module.
2. Work problems (1) and (2) in the Hippo Web Module.
3. The hippo has picked up a river fungus, and now the effective volume of the CSTR stomach compartment is only 0.2 m^3 . The hippo needs 30% conversion to survive. Will the hippo survive?
4. The hippo had to have surgery to remove a blockage. Unfortunately, the surgeon, Dr. No, accidentally reversed the CSTR and the PFR during the operation. **Oops!!** What will be the conversion with the new digestive arrangement? Can the hippo survive?

① - I learned a lot of interesting facts about hippos
- I learned that chemical engineering rxn modeling can be applied to animal digestive tracts
- how to find conversion of food for a fixed V for different type of rxns
- combining CSTR & PFR to model digestion process

②?

③ with given CSTR volume $V = 0.2 \text{ m}^3$
 $X_{\text{out}} = 0.25 \approx 25\%$
Since hippo needs 30% X_{out} , it will not survive

④ if PFR $\rightarrow$ first followed by CSTR
then conversion $\rightarrow 22\%$, hippo needs 30%.
 $\Rightarrow$ will not survive

P2-10_C OEQ (Old Exam Question). The curve shown in Figure 2-1 is typical of a reaction carried out isothermally, and the curve shown in Figure P2-10_C (see Example 11-3) is typical of a gas-solid catalytic exothermic reaction carried out adiabatically.

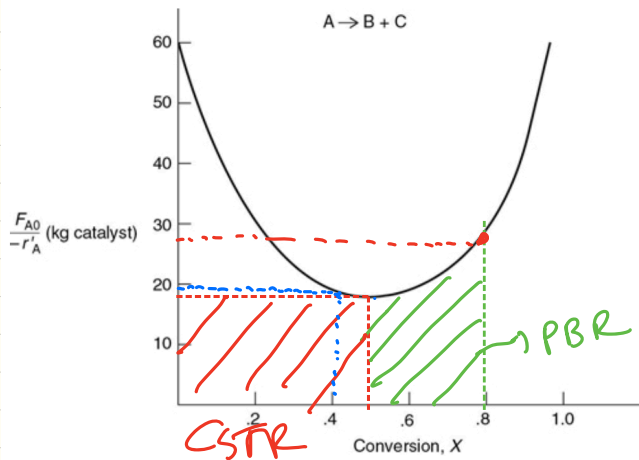


Figure P2-10_C Levenspiel plot for an adiabatic exothermic heterogeneous reaction.

1. Assuming that you have a fluidized CSTR and a PBR containing equal weights of catalyst, how should they be arranged for this adiabatic reaction? Use the smallest amount of catalyst weight to achieve 80% conversion of A.
2. What is the catalyst weight necessary to achieve 80% conversion in a fluidized CSTR? (Ans: $W = 23.2$ kg of catalyst)
3. What fluidized CSTR is necessary to achieve 40% conversion?
4. What PBR weight is necessary to achieve 80% conversion?
5. What PBR weight is necessary to achieve 40% conversion?
6. Plot the rate of reaction and conversion as a function of PBR catalyst weight, W .

Additional information: $F_{A0} = 2$ mol/s.

Equation):

CSTR:

$$W_{CSTR} = \left(\frac{F_{A0}}{-r_A'} \right)_{out} (X_{out} - X_{in})$$

PBR:

$$W_{PBR} = \int_{X_{in}}^{X_{out}} dx \left(\frac{F_{A0}}{-r_A'} \right)$$

① First we use a CSTR until we reach about 50% conversion (minimum point on plot) then we use a PBR from 50% to 80% to minimize catalyst weight needed

$$\begin{aligned} \textcircled{2} \quad W_{CSTR} &= \left(\frac{F_{A0}}{-r_A'(X_{out})} \right) (X_{out} - X_{in}) \\ W_{CSTR} &= (29) (0.8) = 23.2 \text{ kg catalyst} \end{aligned}$$

$$\textcircled{3} \quad W_{CSTR} = \frac{F_{A0}}{-r_A(0.4)} (0.4 - 0) = 20 (0.4) = 8 \text{ kg}$$

$$(4) \quad W_{PBR} = \int_0^{0.8} \frac{F_{A0}}{r_A(x_{out})} dx$$

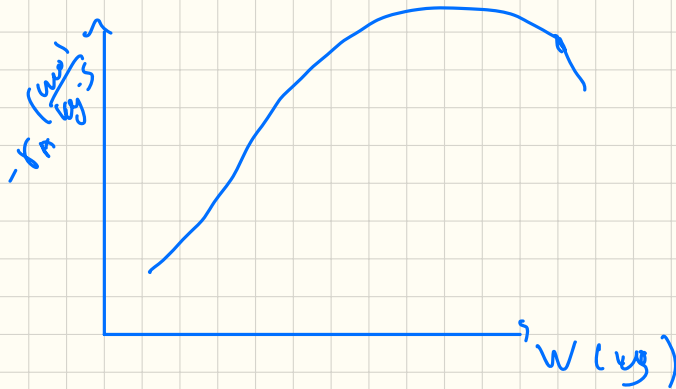
$\Rightarrow$ integrating area under the curve

$$\approx W_{PBR} \approx 22.7 \text{ kg cat.}$$

$$(5) \quad W_{PBR} = \int_0^{0.4} \left(\frac{F_{A0}}{r_A(x_{out})} \right) dx \approx 14.0 \text{ kg cat.}$$

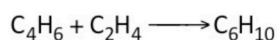
$$(6) \quad F_{A0} = 2 \text{ mol/s}$$

Expected plot:



Q3-2

- 7) If the reaction below is irreversible and elementary as written, which of the following would be appropriate units for the rate constant k ?



A) $(\text{mol}) (\text{dm}^{-3})^{-1} (\text{min})^{-1}$

B) $(\text{mol})^{-1} (\text{dm}^3) (\text{min})^{-1}$

C) $(\text{mol})^{-1} (\text{dm}^3) (\text{min})$

D) $(\text{min})^{-1}$

E) None of the above.

$$-r_A = k C_A C_B$$

$$\left[\frac{\text{mol}}{\text{dm}^3 \text{ min}} \right] = k \left(\frac{\text{mol}}{\text{dm}^3} \right)^2$$

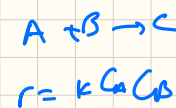
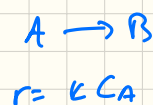
$$k = \frac{\text{dm}^3}{\text{mol} \cdot \text{min}}$$

• This is a good question to use in the exam:

- helps student understand elementary rxns and what the rate law should look like

- helps student get a deeper understanding of the units needed for rate laws and rate constant.

P3-12B Determine the rate law for the reaction described in each of the cases below involving species A, B, and C. The rate laws should be **elementary** as written for reactions that are either of the form $A \rightarrow B$ or $A + B \rightarrow C$.



1. The units of the specific reaction rate are $k = \left[\frac{\text{dm}^3}{\text{mol} \cdot \text{h}} \right]$: Rate Law

$-r = k C_A C_B$ so that $\rightarrow$

$$r [=] \frac{\text{mol}}{\text{dm}^3 \cdot \text{h}}, \quad C = \frac{\text{mol}}{\text{dm}^3}$$

$$r = k C_A C_B = \left[\frac{\text{dm}^3}{\text{mol} \cdot \text{h}} \right] \cdot \left[\frac{\text{mol}}{\text{dm}^3} \right]^2 = \frac{\text{mol}}{\text{dm}^3 \cdot \text{h}}$$

2. The units of the specific reaction rate are $k = \left[\frac{\text{mol}}{\text{kg-cat} \cdot \text{h} (\text{atm})^2} \right]$: Rate Law

$-r = k P_A P_B$

$$r [=] = \frac{\text{mol}}{\text{kg-cat} \cdot \text{h}}$$

3. The units of the specific reaction rate are $k = \left[\frac{1}{\text{h}} \right]$: Rate Law $r = k C_A$

$$r = k P_A P_B = \frac{\text{mol}}{\text{kg-cat} \cdot \text{h} \cdot \text{dm}^2} \cdot (\text{dm})^2$$

4. The units of a nonelementary reaction rate are $k = \left[\frac{\text{mol}}{\text{dm}^3 \cdot \text{h}} \right]$: Rate Law

$r = k \frac{C_A}{1 + C_A}$

3) $r = k C_A = \frac{1}{\text{h}} \cdot \frac{\text{mol}}{\text{dm}^3} \Rightarrow r [=] \frac{\text{mol}}{\text{dm}^3 \cdot \text{h}}$

Rate $\Rightarrow$ mol per unit time per unit volume

4) r has to be $r [=] \frac{\text{mol}}{\text{dm}^3 \cdot \text{h}}$ if $k [=] \frac{\text{mol}}{\text{dm}^3 \cdot \text{h}}$ also

then concentrations must end up being dimensionless i.e. $\Rightarrow r = k \frac{C_A}{1 + C_A}$